

NANDHANA K

Electronics and Communication Engineering | Digital Explorer

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📍 Coimbatore

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Objective

Electronics and Communication Engineering graduate seeking a full-time role to apply skills in embedded systems, programming, and electronics design while contributing to innovative technology solutions and professional growth.

Technical Qualification

Programming : C, JAVA

Hardware : Arduino, ESP32, Raspberry pi, Microcontrollers

Tools : MATLAB, Xilinx, Wokwi, Arduino IDE, VS Code

Core Concepts: Data Structures, OOPs, Basic DBMS, OS

Key Strength:

- Good time management and organizational skills.
- Adaptable to new environments and challenges.
- Skilled in Hardware-Software Integration, Testing and Debugging
- Eagerness to continuously improve technical knowledge

ACADEMIC CREDENTIALS

B.E Electronics And Communication Engineering, Dr. N.G.P Institute of Technology

CGPA: 8.6 (till 5th semester)

HSC : P.S.G. Sarvajana Higher Secondary School - May 2023 | 88.33%

SSLC : P.S.G. Sarvajana Higher Secondary School - March 2021 | 100%

PROJECTS

Smart Farmer Assistance App with IoT Integration 2025-2026

- Developed an IoT-based Smart Farming System using ESP32 to monitor soil moisture and temperature with automatic irrigation control.
- Integrated a mobile application providing farming suggestions, pest alerts, voice guidance and multi-language support.

Design and Optimization of Microstrip Patch Antenna for 2.45 GHz ISM Band 2025-2026

Applications

- Designed a 2.45 GHz microstrip patch antenna using CST Studio Suite on an FR4 substrate for wireless communication applications.
- Optimized the antenna for good impedance matching, low return loss, and acceptable VSWR, suitable for Wi-Fi, Bluetooth, and IoT applications

Smart ESP32 Vehicle Speed Monitoring System with RFID Alerts 2024-2025

- Developed a Smart ESP32 Vehicle System with real-time speed monitoring and display for improved vehicle tracking.
- Integrated RFID-based alert notifications to enhance road safety through instant detection and alerts

Remote Control Vacuum Cleaner

2023-2024

- Developed a Remote Control Vacuum Cleaner integrating hardware and software for wireless operation and smart cleaning.
- Implemented motor drivers, sensors, and remote-control modules for efficient navigation and hands-free floor cleaning.

PROFESSIONAL EXPERIENCE

IoT Intern — KRISH Tech, Coimbatore (June 2025)

- Completed Industrial Practical Training at Krishtec, Coimbatore focused on IoT, embedded systems, and real-time electronics.
- Worked on ESP32 projects, including PWM control and sensor data display.
- Interfaced hardware components such as LEDs, LCDs, buzzers, and dot matrix displays for embedded applications.
- Gained hands-on experience in Wokwi simulation, I2C/SPI communication protocols, and hardware troubleshooting.

Winter Internship — Coimbatore Institute of Technology (Nov 2025)

- Undertook training on Smart Systems and Emerging Technologies with a focus on modern engineering applications.
- Learned fundamentals of embedded systems and real-time applications.
- Enhanced practical understanding of hardware integration and automation concepts.
- Developed knowledge of system design, implementation, and embedded technology workflows.

PROFESSIONAL INTERESTS

- Internet Of Things
- Wireless Communication
- Embedded System

LANGUAGES

- Tamil
- English

ACHIEVEMENTS & CERTIFICATIONS

- Attended the workshop on PCB Designing and Fabrication at Pinnacle 2024 – National Level Technical Symposium, conducted by Coimbatore Institute of Technology (2024–2025).
 - Strengthened knowledge in Cryptography and Network Security through the NPTEL course, achieving a score of 63% (2025–2026).
 - Gained exposure to Water Quality and People's Water Data through the NPTEL course offered by IIT Madras (2025–2026).
 - Demonstrated technical knowledge, teamwork, and problem-solving skills through participation in TNWISE Hackathon 2026.
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Place:

Date:

SIGNATURE